Stefan Busch

82119 Germering, Germany, +49 177 84 46 811, stefan.busch@sb-systec.de
www.linkedin.com/in/stefan-busch-b3027a261

SUMMARY

Senior **Software Developer** and **Architect** with more than **20 years of experience** designing and evolving scalable software systems in technically demanding and regulated environments.

Experienced in **building and maintaining** modular **microservice platforms**, modernizing **legacy enterprise systems**, and implementing **validation-driven** development processes in medical and industrial contexts.

Strong **hands-on** background in .NET, C++, and Python, combined with practical experience in CI/CD, automated testing, and **cross-team collaboration**. Focused on building reliable, maintainable systems through **pragmatic architectural decisions** and close cooperation with developers, testers, and domain experts.

Seeking a senior development or architecture role in **Denmark** within a collaborative and **quality-oriented** engineering environment.

KEY ACHIEVEMENTS

- Architected and evolved a modular microservice ecosystem (100+ modules) in an enterprise environment
- Led migration from monolithic .NET systems to scalable .NET Core REST architecture
- Designed FDA-aligned validation and automated testing strategies in regulated medical software
- Built secure, state-oriented AI multi-agent architecture for aerospace research applications
- Established CI/CD pipelines and quality standards across multi-team environments
- Mentored developers in clean architecture, testing strategies, and engineering best practices

SELECTED FREELANCE WORK EXPERIENCE

09/2015 – 11/2025 **MAN Truck & Bus,** Munich

Senior Software Developer/Architect,

Design and evolution of a scalable modular full-stack microservices platform in an enterprise environment. Responsible for architecture, quality standards, CI/CD enablement, and cross-team technical alignment.

- Designed and evolved a modular microservices ecosystem (100+ modules) across multiple business domains
- Led the migration from legacy .NET monoliths to a REST-based .NET Core architecture, improving system scalability and maintainability
- Established CI/CD pipelines and automated testing strategies, significantly reducing manual deployment effort and increasing release stability
- Integrated distributed components including Elasticsearch, RabbitMQ, and S3-compatible storage solutions
- Conducted architecture reviews and aligned development standards across multiple teams
- Contributed hands-on to backend (.NET Core) and frontend (WPF, Blazor) development in close collaboration with product owners and domain experts

01/2013 – 08/2015 **Pixel,** Munich

Senior Software Developer

Development of a solar system planning application with focus on usability, cross-platform support, and software quality. Responsible for UI/UX design, mobile enablement, and comprehensive test management.

- Implemented UI/UX concepts and mobile enablement
- Designed and executed structured test plans (unit, integration, system tests)
- Test reporting and quality assurance

05/2011 – 12/2012 **Rohde & Schwarz,** Munich

Senior Software Developer

Development of a framework for software-defined radio (SDR) systems with focus on extensibility and automation.

- Designed a domain-specific language (DSL) for SDR system customization
- Implemented C++ code generators based on DSL definitions
- Developed visualization tools for generated system workflows

02/2010 – 04/2011 **T-Systems,** Munich

Software Developer

Design and automation of enterprise-scale BizTalk solution deployments across complex multi-server environments. Responsible for deployment tooling, environment preparation, and post-deployment validation.

- Development of a deployment toolset for large BizTalk solutions (hundreds of artifacts)
- Rollout across ~50 single- and multi-server environments
- Preparation and configuration of target systems
- Conducted integration testing and post-deployment validation

---

SELECTED CONSULTING PROJECTS

02/2025 – 12/2025 **Research, Aerospace, Defense,** Germany

AI System Architect , Prompt Engineer, Context Engineer

Designed and implemented a secure, state-oriented AI system architecture for research-driven aerospace applications with strong focus on validation and deterministic behavior.

- Developed state-oriented AI agents to enhance system effectiveness.
- Orchestrated AI multi-agent systems to facilitate complex interactions.
- Modelled prompts and context for dependable AI behaviour.
- Designed comprehensive system-level AI architecture and orchestration strategies.
- Utilised technology stack with Olllama, .NET Core 10, Blazor, Python.

03/2020 – 09/2023 **Roche,** Penzberg

Test Manager, Senior Software Developer

Design and implementation of validation and test strategies for medical and laboratory software in a highly regulated environment. Focus on automation, cross-technology integration, and regulatory compliance.

- Defined and implemented FDA-aligned test strategies ensuring traceable and compliant validation processes
- Designed a cross-technology test framework that reduced manual testing effort and improved regression stability
- Introduced automated UI and integration testing (Selenium), increasing test coverage and early defect detection
- Supported system validation activities contributing to regulatory approval readiness
- Collaborated closely with developers, quality assurance and domain experts to align technical implementation with regulatory requirements

11/2017 – 02/2019 **ASM Assembly Systems,** Munich

Senior Software Developer, QA Manager

Design and implementation of RPC services for controlling material logistics workflows in PCB assembly automation. Responsible for cross-system testing strategy, execution, automation and operational runbooks.

- Development of RPC services for robotics workflows
- Definition of cross-system test strategy
- Test plan execution and reporting
- Creation of executable cross-system runbooks

MORE PROJECTS www.linkedin.com/in/stefan-busch-b3027a261

SKILLS

Core Architecture & System Design

- Distributed Systems Architecture
- Microservices & Modular Platforms
- Event-Driven Systems
- RESTful & API-First Design
- Domain-Driven Design (DDD)
- Secure & Regulated System Design
- AI System Architecture
- Multi-Agent Systems

Programming & Technology Stack

- .NET Core
- NET Framework
- C#
- C / C++
- Python
- Blazor, Razor
- WPF (MVVM)
- JavaScript

Data & Integration

- PostgreSQL
- SQL Server
- Oracle
- EventStore
- Elasticsearch
- Message Queues (RabbitMQ)
- S3 / MinIO

Engineering

- Architecture Reviews
- Technical Mentoring
- Coding Standards & Best Practices
- Cross-Team Enablement
- Clean Code & SOLID Principles

Quality & Compliance

- Test Strategy & Test Management
- TDD / BDD
- Automated UI & Integration Testing
- Continuous Quality Engineering
- FDA-Aligned Validation Processes
- Shift-Left Security

DevOps & Infrastructure

- CI/CD Pipeline Design
- Docker
- Git
- Linux & Windows Environments
- Deployment Automation

EDUCATION

10/1985 – 08/1990 **Diploma** in Technical Physics, FH Munich

10/1981 – 05/1984 **Journeyman's certificate**, Central Heating & Ventilation Engineering

LANGUAGES

German	English	Danish
C2	C1	A1 (further studies at UCplus Sønderborg ongoing)